CONSOLIDATED APPLICATION-BUILDING PROMPTS



CreditSight

Smart Prompt Guide

Explainable Credit Risk and Predictive Liquidity Intelligence for Finance Teams

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A reusable master prompt and 15 phased prompts covering the complete CreditSight demonstration, its visual refinements, financial calculations, AI assistant, reports and validation.

Reconstructed from the original prompt pack, project discussions and current application. Not a verbatim chat transcript. Synthetic data; illustrative financial assumptions.

Build in controlled phases

Use Prompt 01 first to establish scope. Then apply one phase at a time. Keep existing working features intact and ask for the acceptance check before moving on.

Reusable closing instruction

“Implement only this phase. Preserve the synthetic-data boundary. Explain assumptions, run the relevant checks, and state what is implemented, simulated or deferred.”

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Scope note

Prompts 01–15 consolidate the prototype’s build journey. Prompt 16 is optional production work. Some controls are demonstrations only; the notes on each page prevent them being mistaken for persistent operational workflows.

Master build prompt

Define the product, scope and guardrails before creating screens.

COPY-READY BUILD PROMPT

Act as a finance-product designer and application builder. Build CreditSight: Explainable Credit Risk and Predictive Liquidity Intelligence for Finance Teams. This is the capstone project of Rajan Subramanian, AICA L2 B121.

Create a usable browser-based prototype for a UAE-based, multi-division enterprise. Reporting currency is AED equivalent. Intended audiences are credit analysts, credit managers, collectors, division controllers and CFOs; these are personas, not implemented access roles in demo mode.

Keep the portfolio synthetic and deterministic: 60 customers, four divisions, seven countries and eight sectors. Use one shared generator for customers, exposures, aging, invoices, alerts and promises to pay. Do not introduce real-data storage, login or live financial integrations unless I explicitly request a production migration. Lovable Cloud may power the portfolio assistant only.

Build a standalone branded cover at / and the working dashboard at /dashboard. Include Customers and Customer 360, Onboarding & Scoring, Credit Limits, Monitoring & Alerts, Collections, Provisioning (ECL), Sector & Country, Reports, Data Import, Liquidity Forecast, CFO Stress Test and Portfolio Assistant.

Prioritise explainable calculations, consistent figures, professional exports, readable navigation and honest limitations. Never call a synthetic score a bureau score or a rules-based forecast a validated machine-learning prediction. Build in phases, verify each phase, and preserve working behaviour.

ACCEPTANCE CHECK

Cover and dashboard are separate; every named module is reachable; all business data is synthetic and consistent.

SCOPE & HONESTY NOTE

This is a consolidated reconstruction prompt, not a verbatim historical transcript or evidence that every original brief requirement shipped.

Foundation and synthetic portfolio

Establish one reliable source for every demonstration figure.

COPY-READY BUILD PROMPT

Implement the CreditSight foundation using the existing TanStack Start and React stack. Keep routes under src/routes and shared layouts around the router outlet. Do not switch routing frameworks.

Use a deterministic seeded generator in src/lib/demo-data.ts. Include Automotive Retail, Fleet and Leasing, Parts and Service, and Financial Services. Cover UAE, KSA, Oman, Kuwait, Qatar, Bahrain and Egypt; include eight demo sectors. Generate 60 customer records with IDs, legal-name-style demo names, trade licences, division, country, sector, payment terms, limit, exposure, overdue amounts, aging, score factors and financial ratios.

Use a fixed demonstration snapshot of 25 October 2026, visibly labelled. Construct open invoice records so each customer's invoice principal reconciles exactly to exposure. Include future, partial and missed promises to pay. Treat all money as AED equivalent; do not suggest that stored FX examples are live rates.

Use shared formatters and stable IDs throughout the app. Add search, working navigation, browser-friendly error states and page-specific titles/descriptions. Keep business-record changes local to the demo unless a migration is separately approved.

ACCEPTANCE CHECK

Every customer’s invoice total equals exposure; portfolio totals agree across pages and exports; reloading reproduces the same sample data.

SCOPE & HONESTY NOTE

The original uploaded brief proposed database seeding and about 1,500 invoices. The implemented demo uses a smaller deterministic invoice portfolio; do not present the original target as an achieved fact.

Executive dashboard and Customer 360

Turn a ledger-style portfolio into a scannable management view.

COPY-READY BUILD PROMPT

Build an executive dashboard at /dashboard using the shared demonstration portfolio. Show total receivables, overdue exposure, risk mix, limit utilisation and key alert counts with AED labels and consistent number formatting. Include aging distribution, twelve-month sample trends, top customer exposures and sector/country concentration views. Label illustrative metrics and avoid inventing live trends.

Create a searchable, filterable Customers list and a dedicated Customer 360 page per customer ID. Show division, country, sector, terms, financial ratios, exposure, limit, utilisation, overdue aging, score/rating and factor contributions. Provide relevant links to collections and risk views without implying an action was executed.

Use compact finance-grade tables, right-aligned numeric columns, sticky table headers and consistent risk chips. Make filters and navigation genuinely work. Maintain a visible return to the cover page and keep the last menu entry reachable in shorter windows.

ACCEPTANCE CHECK

Search for a sample customer, open its detail page and reconcile its exposure and score factors back to the list.

SCOPE & HONESTY NOTE

Dashboard evidence is a synthetic snapshot. Do not imply daily synchronisation, actual ledger transactions or permission-based division isolation.

Onboarding and scorecard v1.2

Provide a transparent baseline model for comparison.

COPY-READY BUILD PROMPT

Build Onboarding & Scoring with a selectable v1.2 baseline and v1.3 enhanced model. Keep v1.2 available rather than silently replacing it.

For v1.2 show the eight factor maxima: liquidity 15, leverage 15, profitability 10, payment history 25, trading years 10, sector 10, country 10 and adverse flags 5. Explain each contribution. Use the existing demo formulas: liquidity min(15, current ratio × 8); leverage max(0, 15 − debt/equity × 5); profitability min(10, EBITDA margin percent ÷ 1.5); payment history max(0, 25 − DBT × 0.4). Preserve the remaining implemented v1.2 rules rather than substituting undocumented policy.

Display total score and bands: Low at 75 or above, Medium 55–74, High 35–54, Critical below 35. Expose the inputs used; distinguish calculated score from a real external credit-bureau report. Keep document-upload areas clearly demo-only when they do not actually extract or store documents.

ACCEPTANCE CHECK

Change a ratio or DBT and confirm the factor contribution and score respond; switch between both versions without breaking the form.

SCOPE & HONESTY NOTE

The weights and formulas are illustrative policy assumptions, not independently calibrated underwriting models.

Enhanced scorecard v1.3

Explain data coverage, interpolation and adverse-condition caps.

COPY-READY BUILD PROMPT

Implement v1.3 in a pure scoreCredit function accepting typed ScoreInput. In the form use state for inputs and memoise the calculated result. Display score, risk band, recommended action, factor bars, input explanations, override warnings and a badge showing rounded confidence × 100 as “data coverage”.

Use piecewise-linear scoring curves for liquidity, leverage, profitability, twelve-month days beyond terms and trading years. Preserve factor maxima of 15, 15, 10, 25, 10, 10, 10 and 5. Do not award full credit for missing values. Rescale earned points over the maxima of provided factors; coverage equals provided factor maxima divided by 100. Explain that coverage measures input completeness, not predictive accuracy.

Apply caps, not penalties: negative equity or three or more bounced cheques in twelve months caps score at 34; DBT of 90 or more caps at 54; litigation/ownership change with DBT above 60 caps at 54. Apply the most restrictive applicable cap before rounding and band selection.

Keep v1.3 sector/country maps distinct from portfolio geography. Unlisted country codes use the model's documented fallback. Show model version and identify all curves as requiring calibration.

ACCEPTANCE CHECK

Test 34/35, 54/55 and 74/75 band boundaries, missing-data coverage and each override with concrete inputs.

SCOPE & HONESTY NOTE

v1.3’s selectable country set includes India; the demo portfolio includes Egypt. A UAE company’s official licence lookup is not a credit score.

Credit limits and early-warning alerts

Make credit decisions and risk triggers traceable.

COPY-READY BUILD PROMPT

Build the Credit Limits calculator from the selected sample customer. Show all three workings: 10% of net worth; monthly revenue × payment terms/30 × 1.2 × 15%; annual revenue × 5% × score-tier multiplier. Tier multipliers are 1.0, 0.6, 0.3 and 0.1 for Low, Medium, High and Critical. Recommend the minimum of these methods, rounded to the nearest AED 50,000, and explain the result.

Show current exposure, existing limit and utilisation. Any demo approval or override must be explicitly session-local, with a reason shown in its local log. Do not imply the actual approved limit or an immutable audit record has been updated.

Build a prioritised Monitoring & Alerts view from the generator’s existing rules: limit breach, utilisation above 90%, score deterioration and 90+ overdue exposure for applicable customers. Show exact triggering amounts or score movements, severity, owner and status. Use the implemented thresholds; do not silently adopt thresholds from an earlier aspirational brief.

ACCEPTANCE CHECK

Independently recompute one recommended limit and inspect a breach alert’s exposure versus limit.

SCOPE & HONESTY NOTE

Production value thresholds, multi-level approvals, notifications and permanent audit trails remain future scope.

Collections and ECL provisioning

Connect recovery priorities with illustrative impairment workings.

COPY-READY BUILD PROMPT

Build Collections from overdue sample accounts, ranked by overdue balance. Show customer, aging, exposure and risk; allow local notes and activity entries. Keep invoice promises to pay in the deterministic generator so forecast calculations use the same commitments. Do not claim demo notes send an email, place a call or create a persisted promise.

Build the Provisioning (ECL) workbench with balances in Current, 1–30, 31–60, 61–90 and 90+ buckets. Default loss rates are 0.5%, 2%, 6%, 15% and 45%. Support an editable forward-looking overlay, initially 10%, and display how it changes the result.

Calculate bucket ECL as bucket balance × loss rate × (1 + overlay percent/100). Show total ECL, aging workings and the assumptions used. Distinguish a zero-overlay matrix result from the displayed overlay-adjusted result in reports. Label this an illustrative provision supporting professional assessment, not a posted impairment charge or proof of IFRS 9/Ind AS 109 compliance.

ACCEPTANCE CHECK

Calculate one bucket by hand at both 0% and 10% overlay; verify changing the overlay affects totals consistently.

SCOPE & HONESTY NOTE

Individual PD/LGD/EAD assessments, write-off approvals, actual communications and ledger posting are not implemented production workflows.

Sector intelligence and data import

Provide useful context without overstating data connections.

COPY-READY BUILD PROMPT

Build Sector & Country intelligence using the synthetic portfolio. Show exposures, overdue values and risk context by sector and country. Make the geography and sector totals reconcile to the portfolio. Label risk indices as demonstration assumptions, not current sovereign ratings.

Create a Data Import demonstration that reads a selected CSV in the browser, previews rows, suggests column mappings and lets the user adjust them. Show accepted and rejected row counts with validation reasons. Make it explicit that the current demonstration does not save imported rows to the portfolio.

Provide a sample invoice CSV and a clearly labelled import template whose columns match the screen. Include customer-master and credit-limit data exports with stable customer IDs. Do not claim robust quoted-CSV parsing, XLSX processing, duplicate detection or permanent import history unless those features are separately built and tested.

Add a brief source note: AECB and commercial providers such as Dun & Bradstreet or Creditsafe require verified commercial arrangements. Official UAE licence portals can support legal-name/KYC checks, but do not provide a company credit score. Do not invent a free credit-score API or suggest a live connection exists.

ACCEPTANCE CHECK

Upload a sample CSV, change a mapping and inspect validation; reconcile country/sector exposure sums to the master total.

SCOPE & HONESTY NOTE

Production ingestion, paid bureau access and automated KYC/document extraction require separate delivery and current provider-term checks.

Predictive liquidity forecast

Predict expected receipts from due dates and promises without double counting.

COPY-READY BUILD PROMPT

Create /liquidity with a rules-based, invoice-level expected-receipts forecast. Use the fixed demonstration snapshot and a default 90-day horizon. Expose Base, Optimistic and Conservative scenarios, promise reliability initially 90%, and additional delay.

Use collection probabilities Low 96%, Medium 85%, High 65%, Critical 40%. Optimistic adds eight percentage points and halves DBT; Conservative subtracts fifteen points and multiplies DBT by 1.5. Clamp probability to 0–100%. Due-date timing uses the customer’s adjusted historical DBT plus additional delay. For already overdue invoices, start at the snapshot and apply at least seven days of delay.

For a current/future promise, allocate only min(invoice amount, nonnegative promised amount) to the promise date, adjusted for extra delay. Allocate the remaining principal to the due-date estimate. Promise probability equals risk-adjusted probability × promise reliability. Missed promises must not be treated as future commitments; return their amount to the due-date/overdue allocation.

Calculate expected receipt = allocated principal × probability. Group weekly receipts, showing promise-backed and due-date portions, contractual amounts and cumulative receipts. Add invoice-level evidence and export. Keep calculations in a pure tested module.

ACCEPTANCE CHECK

For a 100,000 invoice with a 40,000 promise, principal allocations total 100,000—not 140,000. Test missed promises and horizon boundaries.

SCOPE & HONESTY NOTE

This estimates receipts, not total treasury liquidity; timing and probabilities are assumptions, not backtested performance.

CFO stress testing and cash gaps

Convert delayed receipts into funding-gap decisions.

COPY-READY BUILD PROMPT

Create /stress-test on top of the exact baseline payment allocations used by liquidity forecasting. Do not create new invoice principal. Add controls for extra collection delay, broken-promise percentage, a selected customer default, opening cash, weekly outflows and minimum cash floor.

Start with AED 5 million opening cash, AED 8 million weekly outflows, AED 2 million minimum cash and a 90-day Base forecast with 90% promise reliability. Shift payment dates by stress delay; multiply promise probabilities by (1 − broken-promises percent/100); set the defaulted customer's probabilities to zero.

For each week, closing cash equals prior cash plus expected receipts less outflows. Prorate the final partial week’s outflows by actual days/7. Funding gap equals max(0, minimum cash − closing cash). Show baseline versus stressed receipts, at-risk receipts, cash paths, peak gap, first breach and closing cash.

Add presets: Base case; Regional slowdown (30-day delay, 40% broken promises); Largest customer defaults (top exposure default, 14-day delay, 20% broken promises); Liquidity crunch (21-day delay, 60% broken promises, outflows +25%); Collections push (zero added delay, no broken promises, outflows −25%). Rank customer priorities by the reduction in in-horizon expected receipts.

ACCEPTANCE CHECK

Run each preset from common starting assumptions; test final-week prorating, zero-probability default and evidence-to-summary reconciliation.

SCOPE & HONESTY NOTE

A preset transforms current cash inputs; do not repeatedly compound outflow changes unnoticed. Weekly balances do not detect intraday shortfalls.

Professional reports and board pack

Deliver management-ready outputs that match the screen.

COPY-READY BUILD PROMPT

Create a Reports centre with the five existing portfolio report definitions and both PDF and CSV downloads. Use shared report definitions so rows, filters and figures reconcile across formats. Do not build unrelated report totals with separate calculations.

Redesign the CFO board pack to export the current scenario in the browser. Include an executive summary; baseline-versus-stress charts; cash-floor breaches; prioritised management actions; assumptions and limitations; weekly reconciliation; and a paginated invoice-evidence appendix. Identify snapshot date, AED equivalent, chosen scenario and synthetic-data status.

Use a consistent print palette derived from global report tokens: white paper, dark ink, teal accent, muted labels and light rules. Use vector text/tables, repeat table headers, avoid splitting rows, wrap long customer names and number every page. Provide readable margins, professional typography and no cropped charts.

Verify exported receipts, opening/closing cash, outflows and peak gap against the displayed scenario. Inspect every PDF page after rendering, not just the cover. Keep PDF and CSV downloads separate and clearly labelled.

ACCEPTANCE CHECK

Change a stress assumption, export the board pack and read the same scenario inputs and totals back from the PDF.

SCOPE & HONESTY NOTE

The report documents an illustrative decision. It is not financial advice, an audited statement or an approval to act.

Cover, themes and cinematic flow

Create the distinctive CreditSight presentation experience.

COPY-READY BUILD PROMPT

Create a standalone cover page with CreditSight and “Credit & receivables intelligence” prominently visible. Use the CreditSight logo and relevant skyline imagery, with readable text and logo contrast in both themes. Show portfolio infographics from shared data, not fabricated marketing statistics. Provide larger, bold dashboard and stress-test calls to action with a short invitation.

Preserve two themes: original professional light and Obsidian Cyber-Glass. Use Sora headings, Manrope body and JetBrains Mono numbers for Cyber-Glass. Keep colours, glass surfaces and shadows in semantic tokens; save theme choice in browser storage and apply it before first paint.

Use a straight animated order-to-cash sequence in this exact order: Customer → Credit → Sales → Liquidity forecast → Collection. Keep KPI cards beside the main cover content; do not replace the strip with the reverted circular experiment.

Coordinate the travelling pulse with stage glow, icon emphasis, larger bold labels and a golden-yellow stage background. Add staggered entrance and a roughly 6.5-second cinematic loop with a brief pause and soft fade. Respect prefers-reduced-motion by showing a static readable sequence. Verify text and stage sizing remain stable during emphasis.

ACCEPTANCE CHECK

Inspect both themes, active pulse and reduced-motion mode; confirm stage order, contrast and zero overlapping labels.

SCOPE & HONESTY NOTE

Visual polish must not obscure synthetic-data labels or interfere with navigation and readability.

Mobile access and navigation

Make the same application usable beyond the desktop.

COPY-READY BUILD PROMPT

Add CreditSight home-screen icons, app name and a web app manifest supporting standalone launch. Request landscape orientation for the installed web app where supported, but do not promise that every phone/browser will lock orientation.

Keep navigation usable in short landscape windows: constrain the sidebar to viewport height, allow its menu area to scroll and keep the final Portfolio Assistant item reachable. Maintain a clickable logo and a visible return-to-cover action from working pages.

Check cover text, KPI cards, charts, tables and forms at desktop and phone sizes. Use controlled horizontal scrolling for wide finance tables without making the entire page overflow. Preserve clear labels and accessible touch targets. If a portrait navigation alternative is needed, implement it explicitly rather than assuming landscape lock makes it unnecessary.

Apply the saved theme consistently after reload. Verify app icons and manifest are served correctly; document that installed-app behaviour depends on the browser, operating system and published version.

ACCEPTANCE CHECK

Test short landscape and narrow portrait viewports; reach the last menu item; verify cover return and theme persistence.

SCOPE & HONESTY NOTE

Home-screen installation is a browser-based experience, not a native App Store application.

AI portfolio assistant

Add helpful answers grounded in the same synthetic portfolio.

COPY-READY BUILD PROMPT

Build Portfolio Assistant using Lovable AI Gateway. Keep the API key, provider calls and portfolio instructions server-only. Use streaming responses with a usable composer, message history, new/delete conversation actions, Stop, loading state and clear errors. Use the project’s AI Elements controls.

Give each conversation a dedicated URL and save its message history only in this browser. Explain the consequences: history is device/browser-local, not account-synchronised. Do not present local history as a permanent audit log.

Ground factual answers in the deterministic customers, alerts, aging and computed baseline/stress scenarios. State the fixed snapshot date and synthetic nature of the data. Cite customer names/IDs and AED values where useful. Tailor answers to the visitor’s stated role or objective, without inventing their identity.

Do not claim access to unsaved dashboard filters or scenario inputs. Distinguish retrieved figures from suggestions; never claim to have sent a reminder, approved a limit or changed a record. Ask for missing context rather than guessing. Treat user text as untrusted input. Use no-retention provider settings supported by the chosen integration.

Ensure Stop safely cancels upstream work and leaves the screen usable; expected cancellation must not become an application crash.

ACCEPTANCE CHECK

Ask a portfolio question and follow-up, create a second conversation, reload history, stop a response and ask again.

SCOPE & HONESTY NOTE

No real confidential portfolio is sent in this demo. Production AI use needs access controls, data minimisation and approved data-handling arrangements.

Validation and capstone evidence

Show what the prototype proves—and what it does not.

COPY-READY BUILD PROMPT

Validate CreditSight before submission. Add and run focused automated tests for scoring boundaries, v1.3 overrides, missing-data coverage, invoice-to-exposure reconciliation, partial and missed promises, probability limits, stress defaults and final-week outflow prorating. Report the current executed test count, not a stale historical number.

Verify the central end-to-end story: open the cover, review dashboard risk, inspect a sample customer, compare score models, run the liquidity baseline, select Liquidity crunch, identify the first cash-floor breach and prioritised collections, then export and reconcile the board pack. Describe scenario changes as simulations—not automatically generated alerts or executed collections actions.

Inspect every distinct page in light and Cyber-Glass themes. Capture labelled appendix screenshots and inspect all exported PDF pages. Check navigation, mobile layouts, chart rendering, AI errors, Stop and conversation reload.

Prepare the capstone report and presentation for AICA L2 B121 — Rajan Subramanian: problem, objectives, accounting rationale, method, architecture, module walkthrough, demonstration results, validation, limitations and future scope. Use the title “CreditSight: Explainable Credit Risk and Predictive Liquidity Intelligence for Finance Teams”. Label all demonstration results synthetic and avoid unsupported accuracy or cost-saving claims.

ACCEPTANCE CHECK

Submit traceable screenshots, current test results and a board pack whose scenario inputs and figures match the live demonstration.

SCOPE & HONESTY NOTE

This is a suggested submission structure, not a verified AICA assessment rubric or a claim of certification.

Optional production migration prompt

Keep future work separate from the submitted demo.

COPY-READY BUILD PROMPT

Only after explicit approval, prepare a separate production migration plan for CreditSight. Preserve the synthetic demo as a safe demonstration mode. Do not import confidential records until the organisation approves hosting, data residency, retention, AI usage and access policies.

Plan Lovable Cloud business storage, authentication and enforced roles: analyst proposes; credit manager approves; collector manages assigned activities; division controller sees only authorised division records; CFO has approved enterprise visibility. Enforce permissions on every read, write, export and AI-grounding path—not just by hiding buttons. Configure supported sign-in providers and verify access isolation.

Design robust CSV/XLSX ingestion with validation, duplicate handling, import history and reconciliation. Add durable approvals and protected audit records, document evidence, notifications and properly approved bureau/KYC connections. Use TanStack server functions for internal application logic and verified server routes for external integrations; do not replace the project architecture with legacy edge-function examples.

Calibrate scorecards and ECL assumptions against approved historical data. Backtest forecast versus actual receipts, measure bias and error by segment, version models, and obtain finance sign-off. Integrate ERP/ledger and cash-outflow sources only after reconciliation tests. Present this as future work, not capability already delivered.

ACCEPTANCE CHECK

Production approval, role-isolation tests, reconciled imports and documented calibration must precede operational use.

SCOPE & HONESTY NOTE

This optional prompt is intentionally outside the current capstone prototype scope.